Scenario #1: 58 YOM w/ Heat Stroke

Section	Scenario
Scenario Set Up	Equipment: stretcher, stair chair, ice packs, wet sheets, fan, rectal thermometer, nonrebreather, suction, pulse oximeter, BP cuff, glucometer, IV kit with 1 L normal saline, cardiac monitor, cooler of ice water if available, prop bottle of hydrochlorothiazide 25 mg. Background: 58-year-old roofer in 99°F heat, no shade. He collapsed on a flat roof after working since 8 AM on coffee and energy drinks. PROCTOR, patient: Short, slurred answers. Know your name, not your location. Say you are "so hot." If cooling has not started 5 minutes after assessment begins, become unresponsive and have a 30-second seizure. If cooling starts promptly, become clearer. PROCTOR, Luis (proctor 2): Calm but scared. Answer SAMPLE questions for the patient. Hand over the pill bottle when asked.
Dispatch	Priority 1 to a 58-year-old male, confused and vomiting after collapsing on a roof in the heat. Two-story commercial building, flat roof, rear stairwell.
Scene Size Up	Sitting against a wall in direct sun, soaked shirt, vomit on chin. Roof edge is a fall risk. Main threats: heat-related altered mental status and vomit in the airway.
Pertinent Primary Assessment Findings	AVPU: V, A&O x1 (name only). A: open but gurgling; suction needed. B: rapid and shallow, RR 26; lungs clear; SpO2 94% on room air. C: skin hot, flushed, damp; pulse 128, strong; capillary refill 2 seconds.
Pertinent Secondary Assessment Findings	S: headache, dizziness, nausea, vomiting twice, calf cramps. GCS 13. A: none known. M: hydrochlorothiazide 25 mg daily. P: high blood pressure, no prior heat illness. L: coffee at 6 AM, energy drink at 10 AM, no water since then. E: headache, then collapsed and vomited after hours of work in the heat.

Vitals	Initial: BP 96/58, HR 128, RR 26, SpO2 94% on room air, BGL 104 mg/dL, rectal temp 105.6°F (40.9°C). 15 minutes into cooling: BP 104/62, HR 112, RR 22, SpO2 97% on nonrebreather, temp 102.8°F (39.3°C), A&O x2.
Treatments	Do first: (1) Move to shade or the air-conditioned ambulance; remove excess clothing. (2) Cool now: ice-water immersion if available, otherwise ice packs to neck, armpits, and groin with wet sheets and a fan. Check temp every 5 minutes. (3) Suction vomit; recovery position if vomiting continues. (4) Nonrebreather at 15 LPM. (5) IV and 1 L normal saline. (6) Monitor; sinus tachycardia expected. Also: no acetaminophen or ibuprofen; nothing by mouth while confused; transport urgently and keep cooling en route.
Key Points	Heat stroke means core temperature of 104°F (40°C) or higher with confusion, agitation, seizure, or coma. Start cooling during the primary assessment. Stop active cooling at about 102°F (39°C) and watch for rebound. Keep the crew and patient away from the roof edge. Check your local protocol for cooling endpoints and transport criteria.
Bonus Questions	1. How does heat stroke differ from heat exhaustion? (Brain dysfunction with a core temperature of 104°F or higher.) 2. Why is acetaminophen not given? (Heat stroke does not raise the temperature set point.) 3. Why does his hydrochlorothiazide matter? (It reduces fluid and salt reserves.)

Scenario #2: 28 YOF Pulled From Lake Meritt

Section	Scenario
Scenario Set Up	Equipment: stretcher, nonrebreather mask and O2 tank, BVM, portable suction, pulse oximeter, BP cuff, stethoscope, glucometer, tympanic thermometer, towels and blankets, gloves and eye protection. Background: 28 YOF ~62kg swimming ~30 yards out into Lake Merritt on a Saturday afternoon. Water is ~60F. While she was resting on a rock in the middle of the lake, her friend saw her slip off the rock into the water around 1405, 15 minutes before EMS arrival per her friend. Patient panicked in the cold water, gasped, and went under water. The on-duty lifeguard reached her after around a minute of being under, and she was limp but coughing when she resurfaced. On the shore she coughed up and vomited lake water. Her friend called 911 at 1410 while the lifeguard radioed park staff. There are many beachgoers crowding around, and the lifeguard is trying to keep them back. Proctor 1 is playing the patient, sitting upright leaning forward, shivering, coughing, speaking only a few words at a time, mentioning: “I can’t breathe.” If O2 is not given within a few minutes of patient contact, SpO2 drops to 82% and patient becomes more confused. If O2 is given, speak more easily within 5 minutes and say patient “feels better” and wants to go home. En route, patient gets short of breath again with a frothy cough: “I can’t breathe again…” If an EMT tries to walk you to the ambulance, nearly faint. If given anything by mouth, vomit. Proctor 2 is playing the friend and on occasion possibly the lifeguard. Friend is worried but helpful. Answer SAMPLE questions for pt, she has accurate information. Lifeguard mentions submersion was for about 1 minute, no diving, no head strike, no CPR was needed.
Dispatch	28 YOF near-drowning, caller states patient is breathing, coughing, lifeguard is on scene. EMS on scene at 1420.
Scene Size Up	Pt already out of water, sitting on towel in tripod position, wet swimsuit, shivering, lips slightly blue. Vomit on her chin, coughing. A lifeguard and

	her friend by her, with a crowd nearby. Scene safety: BSI including eye protection for vomit, move patient up the beach away from the water's edge, ask the lifeguard to keep the crowd back. Request ALS. Environment safe aside from crowds. MOI/NOI: medical, drowning main C/C, slid off rock with no diving, head strike or trauma so no SMR needed. Hypoxia from aspirated water is the immediate life threat. Coughing, crackles, apparent cyanosis. Rapid pulse, looks sick, priority patient. Start high-flow O2 immediately.
Pertinent Primary Assessment Findings	AVPU: A&Ox3 (missed time) GCS 14 (E4 V4 M6) A: Patent, but at risk: frequent coughing, vomit on chin, suction mouth immediately and wipe vomit. B: RR 30, labored and shallow, bilateral. Tripod position, accessory muscle use, speaking in only a few words, crackles at both bases, SpO2 @ 86% on RA. C: Radial pulse 118, strong and regular frequency. Skin pale, cool and wet from lake, lips slightly cyanotic; cap refill is 3 seconds (cold skin), no bleeding.
Pertinent Secondary Assessment Findings	S: Shortness of breath, persistent cough, chest tightness, nausea, lightheaded, chills. (O) Onset after being pulled from water. (P) Worse when laying down. (Q) Heavy chest, can't get a full breath. (R) Chest w/o radiation. (S) 8/10 difficulty breathing. (T) 15m ago, getting worse. A: No known drug allergies. M: None. P: Mild childhood asthma, no inhaler for years, no further medical history. L: Sandwich/lemonade ~1200. E: Resting on a rock ~30 yards out, slipped into cold water at ~1405. Panicked, submerged ~1 minute, pulled out by the lifeguard, coughed up and

	vomited lake water. BGL is 102, No trauma, Temp 96.8F, mildly cool, Denies alcohol and drugs, No seizure history. Focused exam: lips slightly cyanotic, pupils PERL, trachea midline, no JVD, lungs same as in primary, abdomen soft, PMS intact ×4, no signs of trauma, shivering.
Vitals	Initial (1422, RA): BP 118/76, HR 118, RR 30 and labored, BGL 102, SpO2 86% RA, GCS 14, Tympanic Temp 96.8F/36.0C, Cap Refill 3s, crackles at bases, pupils PERL. After O2 (1429, NRB @ 15 LPM): BP 120/78, HR 104, RR 22, SpO2 95%, GCS 15, A&Ox4, crackles still at bases, lips pink, Cap Refill 2s after drying and blankets. En route (1444): BP 122/80, HR 116, RR 28, SpO_2 89% on NRB, GCS 15 but anxious, crackles spreading up to mid-lung fields, frothy cough, pale. Later, can explain hypoxia (SpO2 86%) as why pt is not A&Ox4 or 15 GCS.
Treatments	1. Move away from water's edge, sit upright, suction vomit. 2. NRB 15 LPM, BVM if becomes inadequate. 3. Remove wet clothing, dry patient, cover with blankets to stop further heat loss. 4. Reassess vitals every 5m. 5. Keep pt from walking, carry to stretcher, semi-Fowler's. 6. Emergency transport to the closest appropriate hospital. 7. Recognize worsening conditions, keep upright, try to calm down patient, code 3, radio receiving facility. - Less immediate: raise ambulance temp, BGL and take temp after more obvious things. - Do NOT give anything by mouth, apply SMR, or immediately accept a refusal when she feels better.

Key Points	- Sudden cold water causes a gasp reflex, leading to aspiration. - Crackles = aspirated water, can get worse hours later (secondary drowning). - Drowning = hypoxia: oxygenate and ventilate first. - Anyone symptomatic after submersion is transported. - No SMR without a diving or trauma mechanism.
Bonus Questions	1. Why did she worsen en route? (Aspirated water washes out surfactant, delayed drowning.) 2. What ALS intervention can help when her condition was worsening? (CPAP) 3. Why remove wet clothing? (Prevents hypothermia, which worsens perfusion.) 4. Why no SMR? (No diving, head strike or trauma.) 5. She feels fine; why transport? (Delayed drowning can appear hours later.)

Scenario #3: 24 YOF With Eye Pain

Section	Scenario Details
Scenario Set Up	Equipment: EMT bag, gloves, eye protection, oxygen, BVM, pulse oximeter, BP cuff, saline or clean water, and gauze. Proctor: You are a 24-year-old woman at work. Something got into your eyes, and they are burning badly. You keep them shut because they hurt too much to open. **You are wearing contact lenses**. Don't tell the EMTs what got into your eyes unless they ask you questions.
Dispatch	Respond to a 24-year-old female complaining of severe eye pain after something got into her eyes at work.
Scene Size Up	The patient is sitting in a chair, crying and covering her eyes with her hands. She keeps saying, “My eyes hurt so bad. I can't see!” A coworker is standing nearby and looks worried. There is some spilled material on the floor and a few containers on a nearby workbench. EMTs should check scene safety and use appropriate PPE before approaching.
Pertinent Primary Assessment Findings	AVPU: Alert. A: Airway patent; speaking normally. B: Breathing slightly fast because she is in pain, but adequate. C: Radial pulse present; skin slightly pale and clammy. The patient is crying and holding both eyes shut. If EMTs ask her to describe the pain, she says both eyes are burning and her vision is blurry. She is very sensitive to light.
Pertinent Secondary Assessment Findings	S: Severe burning pain, tearing, redness, and blurry vision in both eyes. A: No known allergies. M: Takes birth control. P: No significant medical history. L: Ate lunch about an hour ago. E: If EMTs ask what happened, the patient explains that she was working when some material splashed into her eyes. If they ask what the material was, she says she isn't sure. If they ask what kind of material was involved or investigate the work area, her coworker explains that **both a dry powder and a liquid cleaning chemical** got into her eyes. The dry powder landed first, followed by the liquid. The patient rubbed her eyes immediately afterward. **If EMTs ask about contact lenses, she says she is wearing them in both eyes**. She has not rinsed her

	eyes yet.
Vitals	Initially: BP 126/82, HR 106, RR 22, SpO_2 99%. If EMTs delay treatment: BP 140/88, HR 120, RR 28, SpO_2 98%. The patient becomes more distressed and says her vision is getting worse. After appropriate treatment: She may feel some relief after irrigation, but pain, redness, and blurry vision may remain.
Treatments	● Ensure scene safety and use appropriate PPE. ● Assess the patient and recognize the possibility of a chemical eye injury. ● First brush away any remaining dry chemical from the skin around the eyes before starting irrigation ● Do not delay irrigation if no dry material remains or it cannot be safely removed promptly. ● Once loose dry material is removed, immediately irrigate the eyes with copious saline or clean water ● Gently hold the eyelids open as tolerated. ● Removing contact lenses during irrigation. ● Arrange prompt transport and reassess frequently.
Key Points	Dry chemicals should be brushed away before irrigation because water can activate or spread some substances. However, irrigation must not be unnecessarily delayed. Contact lenses can trap chemicals against the eye, so EMTs should follow their protocol for removing them while irrigating.
Bonus Questions	1. Why should you brush away dry chemicals before irrigating? (could activate more chemicals if rinsed first) 2.How would you handle her contact lenses? (store them and give to hospital to analyze substance if needed)